CSS183-2 SA01 - Quality Assurance Report
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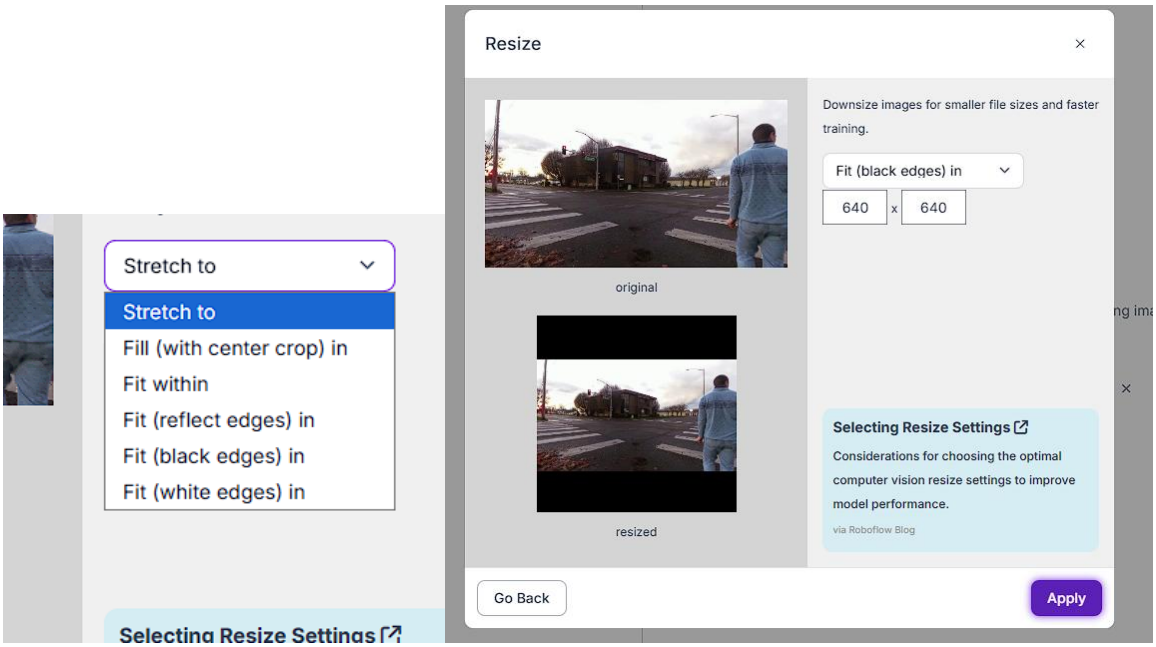
I. Introduction

This report explains the quality assurance steps applied during the preparation of the urban traffic object detection dataset. The purpose of this report is to show how the dataset was checked, standardized, and improved to make sure it is reliable and suitable for machine learning training.

Quality assurance focuses on making sure the images are clear, consistent in size, properly labeled, and balanced enough for model training.

II. Frame Size Standardization



All extracted image frames were resized to 640 × 640 pixels before creating the final dataset version. The resizing option used was Fit (black edges) in the tool Roboflow to avoid stretching or distorting the images. This method preserves the image's original proportions and adds black borders only when it is necessary. By standardizing all images to the same size, the dataset becomes more consistent and easier for models like YOLO to process.

III. Normalization Process

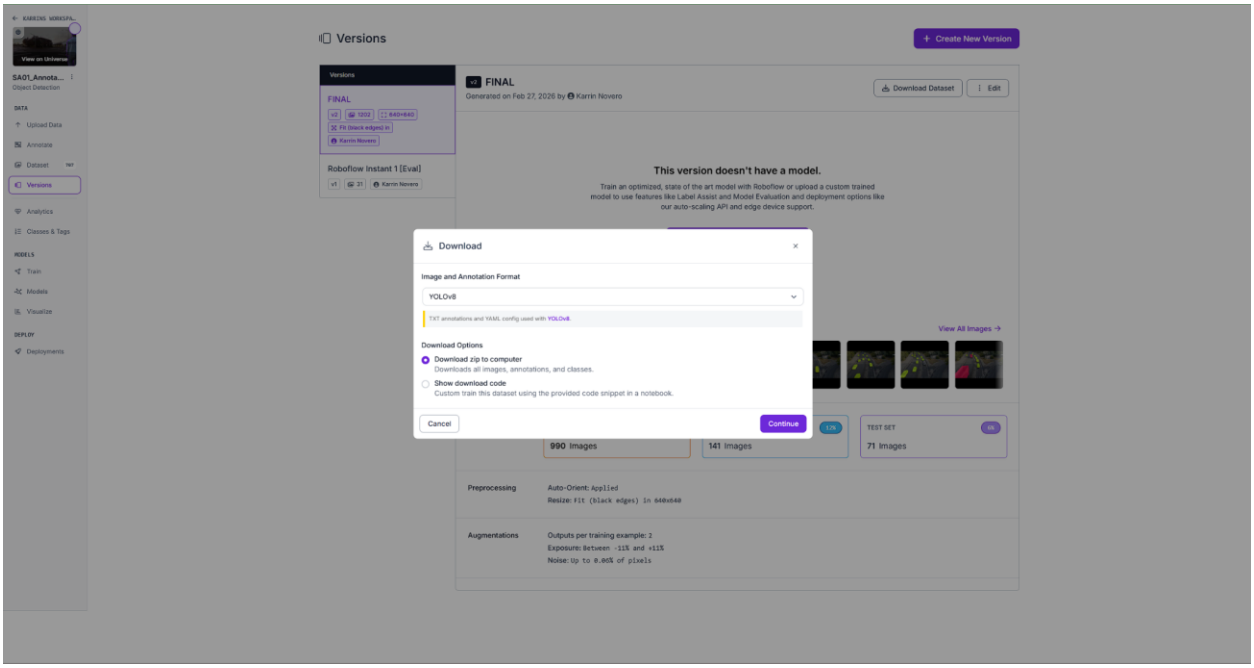
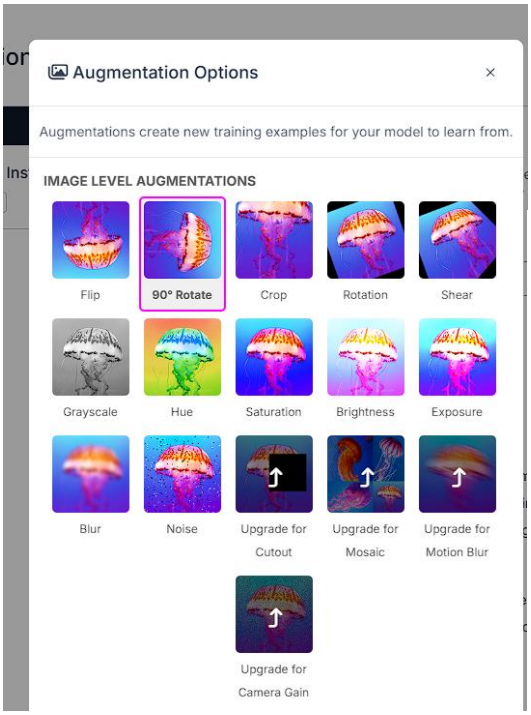
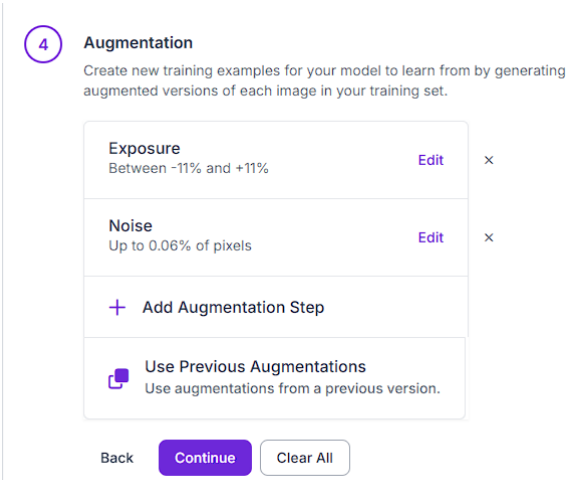


Image normalization was applied automatically when exporting the dataset in YOLO format. This process scales pixel values from their original range (0–255) to a smaller range (0–1).

Normalization helps stabilize the training process and allows the model to learn more efficiently. Since this was handled automatically by the preprocessing pipeline, all images were normalized consistently.

IV. Data Augmentation Configuration

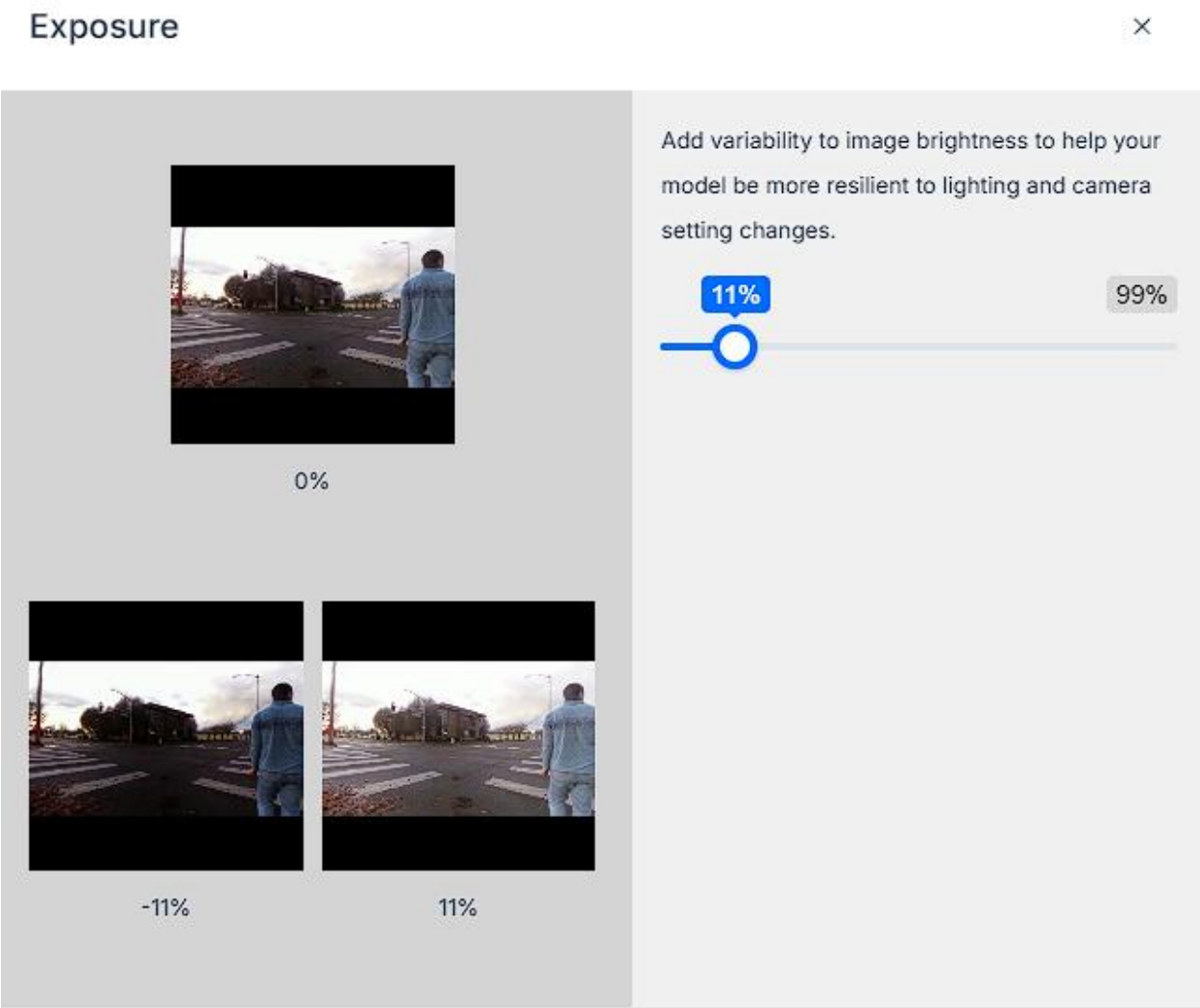




To improve the quality and robustness of the dataset, light augmentation techniques were applied. These included:

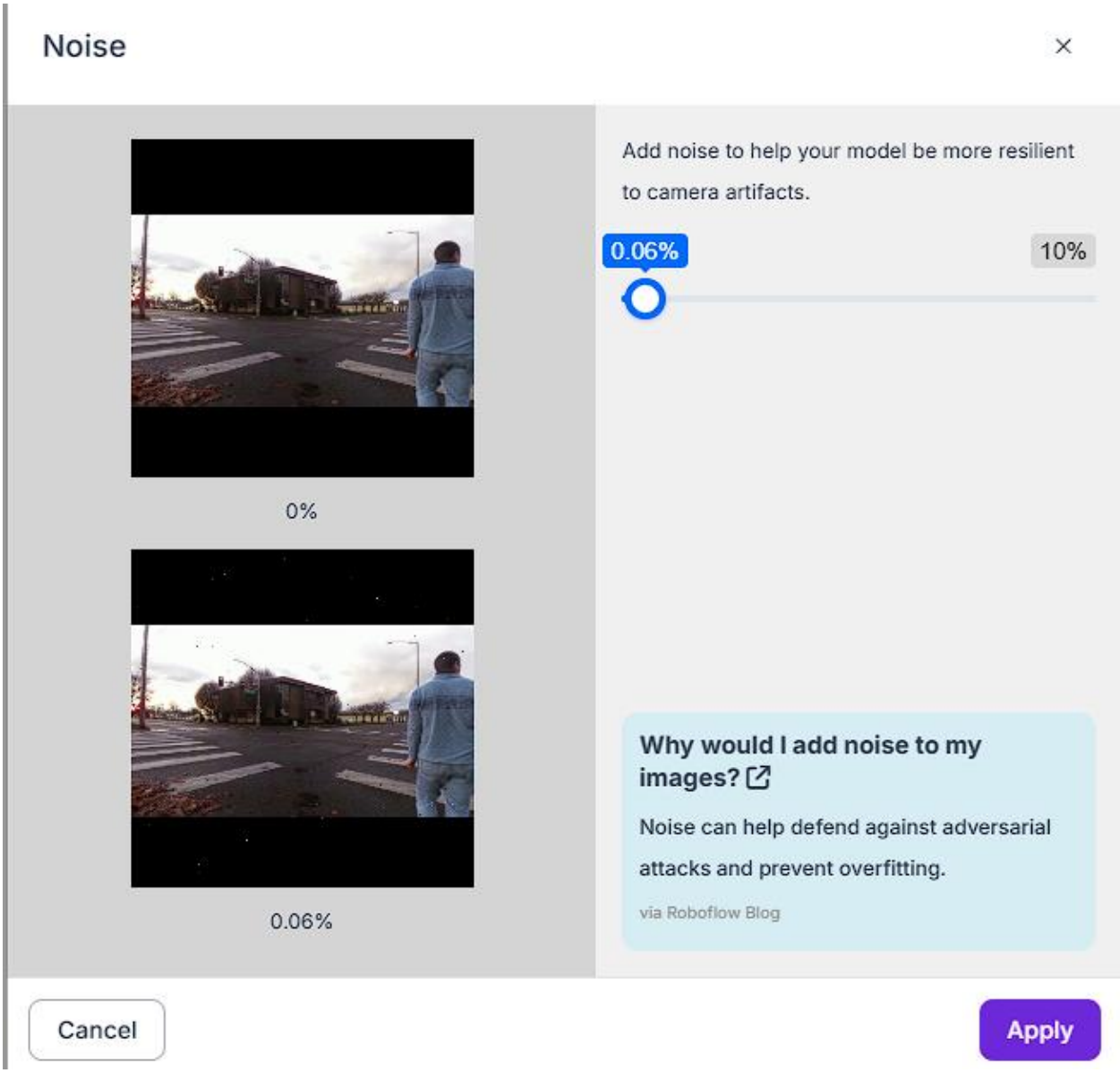
- Exposure adjustment between -11% and +11%
- Noise addition up to 0.06%

These augmentations simulate small lighting changes and realistic camera noise. The settings were kept moderate to avoid unrealistic distortions. After augmentation, the total number of images increased from 707 original annotated images to 1,202 images. This helps improve model generalization without collecting new raw data.



V. Noise Reduction Considerations

During the annotation review stage, blurry or unclear frames were manually checked. Frames with excessive motion blur or poor visibility were either corrected or excluded. Instead of applying heavy noise removal filters, the dataset preserves natural environmental conditions from real-world traffic scenes. This helps the model learn from realistic data.



VI. Motion Stability and Frame Sampling

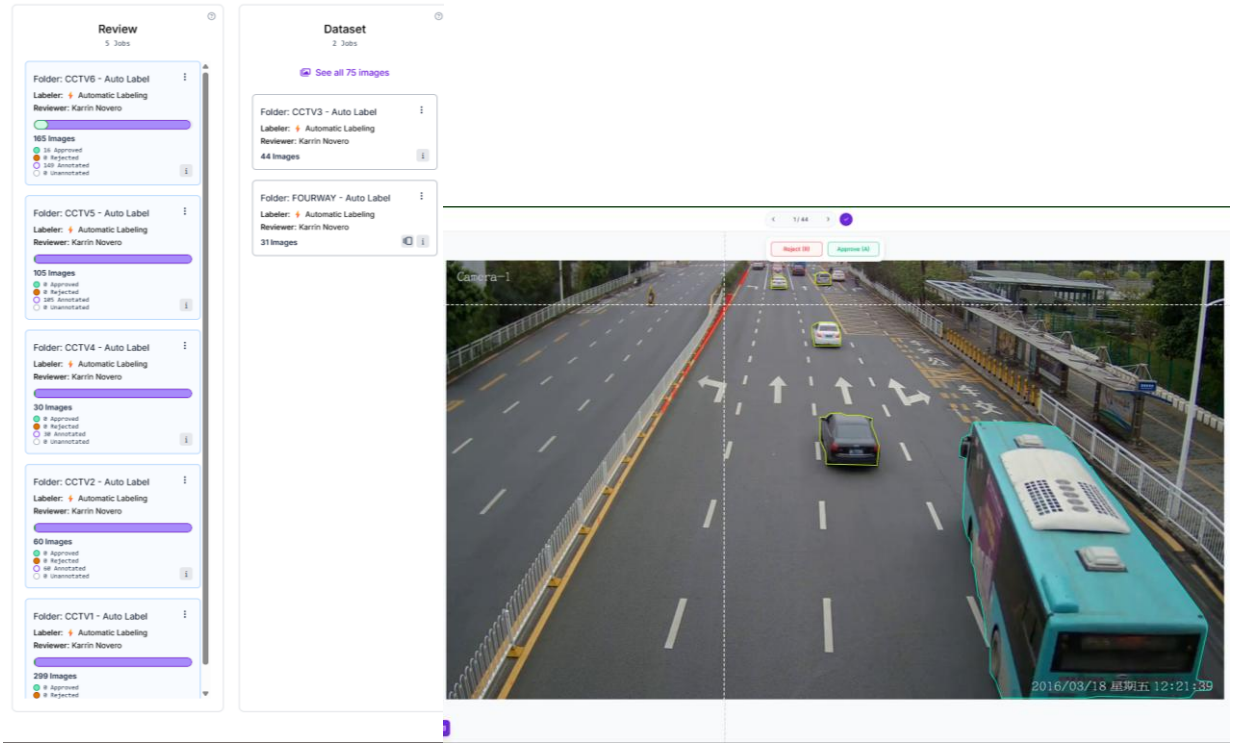
Frames were extracted at a rate of 1 frame per second (1 FPS). This was done to reduce repeated or nearly identical frames from the videos.

Sampling at 1 FPS ensures:

- Better variation between frames
- Reduced redundancy
- More efficient dataset size

This approach also helps minimize motion blur and unnecessary duplication.

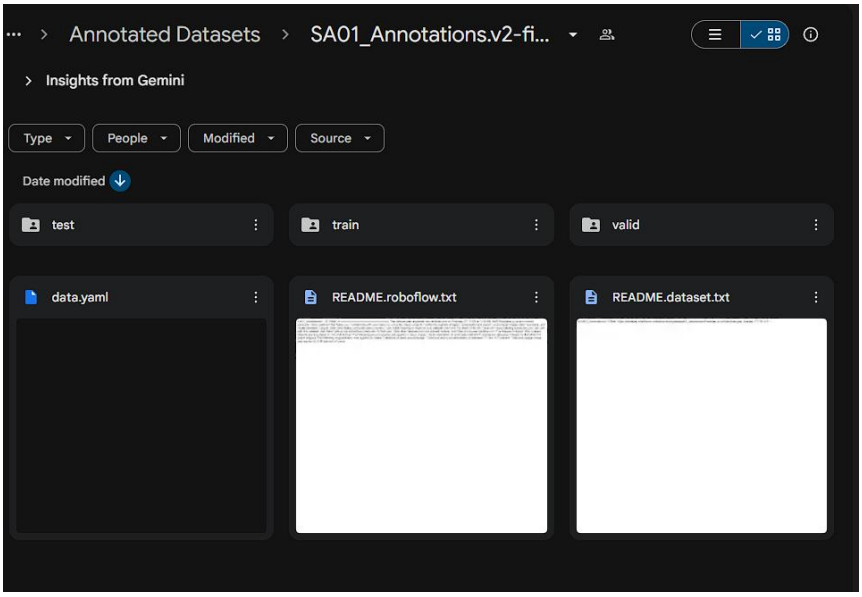
VII. Annotation Verification

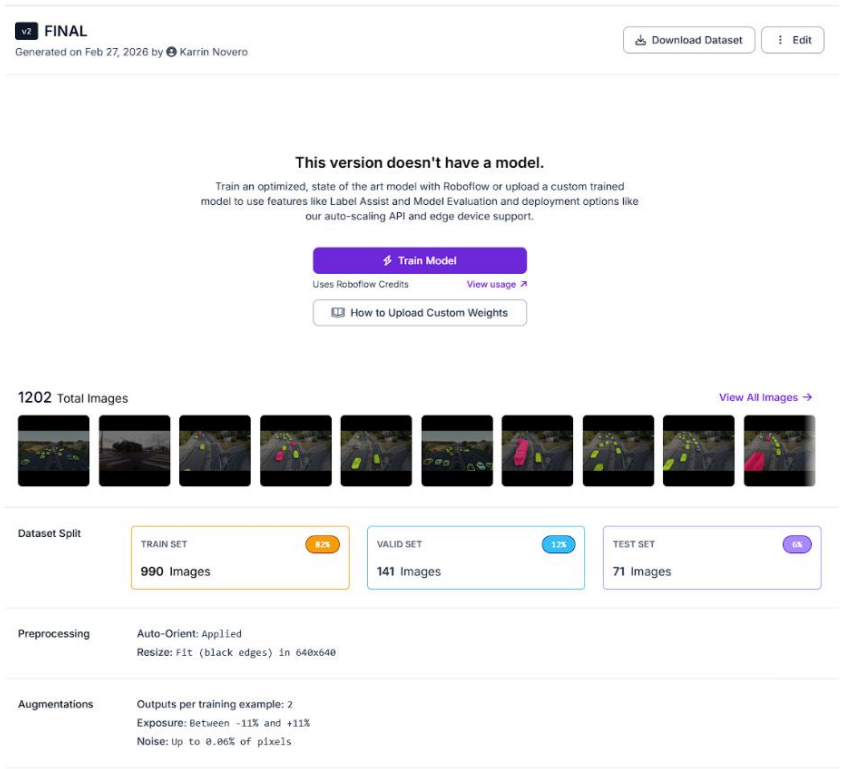


Although auto-labeling was used, the SAM 3 model from RoboFlow, every image was manually reviewed. Incorrect detections were rejected or adjusted. Only approved images were included in the final dataset. The 1000-image dataset that was extracted during the frame extraction was reduced to around 700 avatar annotations and reviews.

This manual review step helps improve annotation accuracy and ensures the dataset is reliable.

VIII. Annotation Verification





The final dataset contains only reviewed and approved images. All frames follow the same resizing configuration, preprocessing settings, and annotation format. No unannotated or incomplete images were included in the final version.

The dataset was also properly split into training, validation, and testing sets to ensure fair model evaluation. These quality control steps ensure that the dataset is consistent, reliable, and ready for machine learning model training.

Reference

https://docs.roboflow.com/annotate/ai-labeling/automated-annotation-with-autodistill
https://www.youtube.com/watch?v=SDV6Gz0suAk
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